RobustGaSP: Marginal posterior mode estimate vs maximum likelikehood estimation

2D test example

Using the branin function

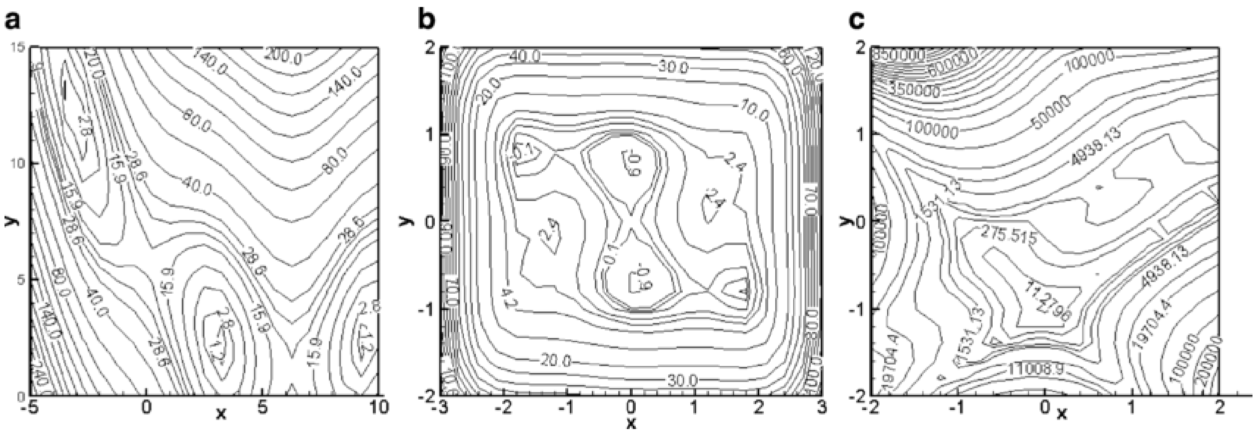


Figure 1: Fig. 2 from Ensemble of surrogates - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/Contour-plots-of-two-variable-test-functions-a-Branin-Hoo-function-b-Camelback_fig9_227093096 (accessed 4 Dec 2025). Contour plots of two variable test functions. (a) Branin-Hoo function, (b) Camelback function, (c) Goldstein-Price

We have a 2-dimensional input, one output, and 20 noise-free observations, obtained using the branin function:

$$x_1 = \{0.76, 0.59, 0.22, 0.84, 0.66, 0.43, 0.11, 0.33, 0.38, 0.95, 0.04, 0.72, 0.5, 0.07, 0.65, 0.16, 0.47, 0.89, 0.99, 0.28\}$$

$$x_2 = \{0.8, 0.27, 0.13, 0.44, 0.81, 0.69, 0.47, 0.87, 0.38, 0.3, 0.65, 0.21, 0.93, 0.17, 0.5, 0.97, 0.02, 0.7, 0.06, 0.57\}$$

$$y = \{139.51, 8.09, 58.93, 40.77, 130.14, 52.37, 33.72, 68.25, 17.53, 6.85, 37.92, 22.37, 128.36, 141.13, 48, 12.85, 15.62, 86.63, 4.18, 18$$

Going forward, we will assume a linear mean function of the form

$$m(x) = \beta_0 + \beta_1 x_1 + \beta_2 x_2.$$

In `RobustGaSP`'s `rgasp()` function, this involves specifying the `trend` matrix H:

```
##       [,1]       [,2]       [,3]
##  [1,]    1 0.76017081 0.79702622
##  [2,]    1 0.58949169 0.27230169
##  [3,]    1 0.22460276 0.12644554
##  [4,]    1 0.84317681 0.44017462
##  [5,]    1 0.66316922 0.80562301
```

```
##  [6,]    1 0.42940075 0.69219925
##  [7,]    1 0.10805479 0.47170505
##  [8,]    1 0.33240879 0.86647123
##  [9,]    1 0.38422386 0.37532350
## [10,]    1 0.94582516 0.30361170
## [11,]    1 0.04383727 0.64642554
## [12,]    1 0.71573814 0.20533548
## [13,]    1 0.50320850 0.93374813
## [14,]    1 0.07028631 0.16755243
## [15,]    1 0.64520444 0.50115916
## [16,]    1 0.15654733 0.96509599
## [17,]    1 0.46585037 0.01688884
## [18,]    1 0.88744150 0.70093504
## [19,]    1 0.99196165 0.06426144
## [20,]    1 0.28021070 0.56665670
```

as well as setting `zero.mean` equal to `"No"`. The Gauss (squared exponential) *and* Matern 5/2 kernels will be are compared for both estimation methods. The output from these four `rgasp` calls is shown below, first using maximum likelihood estimation for the Gauss and Matern 5/2 kernels respectively, and second using marginal posterior mode estimation for the Gauss and Matern 5/2 kernels respectively.

```
## The upper bounds of the range parameters are Inf Inf
## The initial values of range parameters are 0.005938449 0.005939078
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  -104.4821
##  Optimized range parameters are 0.005938449 0.005939078
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 0.1896249 0.1896414
## Start of the optimization  2  :
## The number of iterations is  22
##  The value of the  profile likelihood  function is  -93.42977
##  Optimized range parameters are 0.3566023 1.357531
##  Optimized nugget parameter is 0
##  Convergence:  TRUE

## The upper bounds of the range parameters are Inf Inf
## The initial values of range parameters are 0.005938449 0.005939078
## Start of the optimization  1  :
## The number of iterations is  1
##  The value of the  profile likelihood  function is  -104.4821
##  Optimized range parameters are 0.005938449 0.005939078
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 0.1896249 0.1896414
## Start of the optimization  2  :
## The number of iterations is  10
##  The value of the  profile likelihood  function is  -93.13991
##  Optimized range parameters are 0.9203975 2.629771
##  Optimized nugget parameter is 0
##  Convergence:  TRUE

## The upper bounds of the range parameters are Inf Inf
```

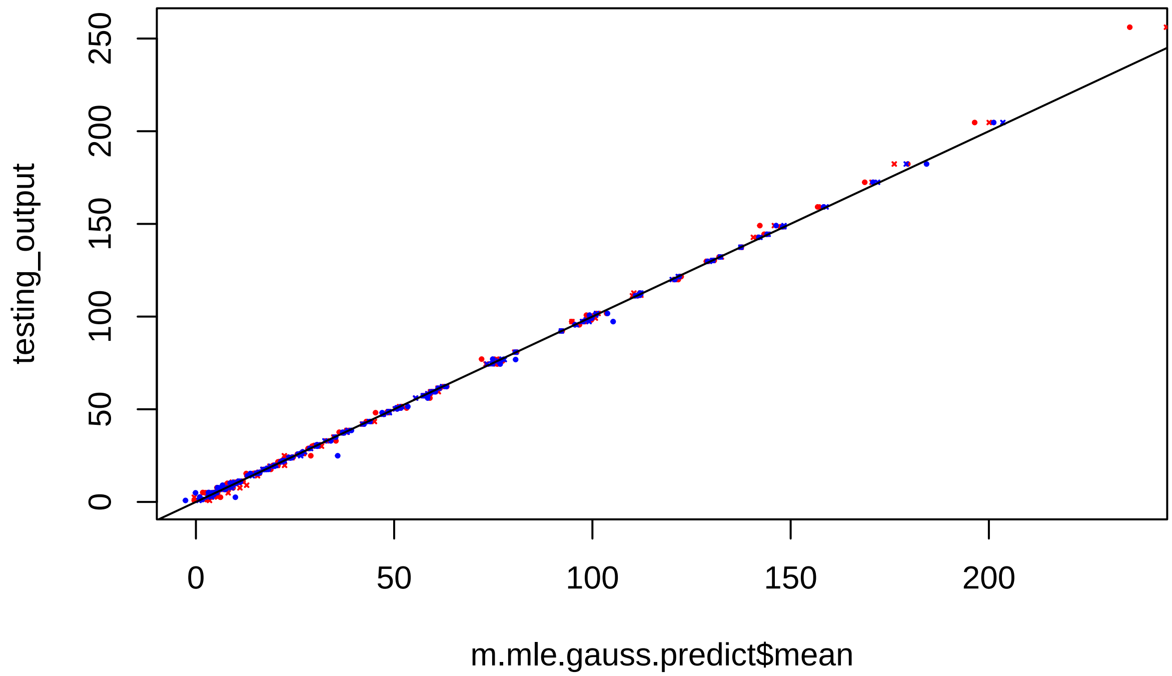
```
## The initial values of range parameters are 0.005938449 0.005939078
## Start of the optimization  1  :
## The number of iterations is  31
##  The value of the  marginal posterior  function is  -75.01643
##  Optimized range parameters are 0.4257828 2.542223
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 0.1896249 0.1896414
## Start of the optimization  2  :
## The number of iterations is  15
##  The value of the  marginal posterior  function is  -74.88994
##  Optimized range parameters are 0.4187454 2.212362
##  Optimized nugget parameter is 0
##  Convergence:  TRUE

## The upper bounds of the range parameters are Inf Inf
## The initial values of range parameters are 0.005938449 0.005939078
## Start of the optimization  1  :
## The number of iterations is  37
##  The value of the  marginal posterior  function is  -116.0005
##  Optimized range parameters are 125081.6 125030.9
##  Optimized nugget parameter is 0
##  Convergence:  TRUE
## The initial values of range parameters are 0.1896249 0.1896414
## Start of the optimization  2  :
## The number of iterations is  30
##  The value of the  marginal posterior  function is  -69.19534
##  Optimized range parameters are 2.493387 8.56707
##  Optimized nugget parameter is 0
##  Convergence:  FALSE
```

Various checks are performed (as in the example from `RobustGaSP`'s Help page):

Table 1: RobustGaSP manual's suite of checks

Estimate method	Kernel	Prop. w/n 95% CI	Out-of-sample MSE	Mean width of 95% CI
Maximum likelihood	Gauss	0.91	8	6.1615
Maximum likelihood	Matern 5/2	1	3	8.1496
Marginal post. mode	Gauss	0.96	5	4.5672
Marginal post. mode	Matern 5/2	1	0.4	6.345



Finally, in preparation for the real world data where we cannot perform out-of-sample validation, we'll perform leave-one-out validation: for each of the observations, fitting a GP using all observations except this one, then using the fitted emulator to predict the output at the excluded observation's input setting. The 95% credible interval for each prediction is checked against the actual output value, and the proportion of these CI's that contain the respective observation is found. The results are shown in Tables 2-5 below.

Table 2: Maximum likelihood with Gauss kernel (65% validation)

actual	loo_prediction	lower95	upper95	validated
139.51	122.65	108.55	136.75	0
8.09	12.68	8.68	16.68	0
58.93	80.12	61.81	98.43	0
40.77	45.91	40.61	51.22	1
130.14	132.25	128.43	136.08	1
52.37	63.65	53.33	73.97	0
33.72	46.52	33.62	59.43	1
68.25	70.43	60.84	80.02	1
17.53	12.96	4.76	21.16	1
6.85	4.23	-3.47	11.94	1
37.92	48.10	24.99	71.21	1
22.37	9.63	1.71	17.55	0
128.36	120.81	111.17	130.45	1
141.13	80.08	52.48	107.68	0
48.00	44.55	40.47	48.63	1
12.85	-0.37	-28.34	27.61	1
15.62	5.96	-12.64	24.56	1
86.63	121.82	89.02	154.63	0
4.18	6.60	-13.93	27.13	1
18.04	13.76	5.35	22.17	1

Table 3: Maximum likelihood with Matern 5/2 kernel (65% validation)

actual	loo_prediction	lower95	upper95	validated
139.51	122.65	108.55	136.75	0
8.09	12.68	8.68	16.68	0
58.93	80.12	61.81	98.43	0
40.77	45.91	40.61	51.22	1
130.14	132.25	128.43	136.08	1
52.37	63.65	53.33	73.97	0
33.72	46.52	33.62	59.43	1
68.25	70.43	60.84	80.02	1
17.53	12.96	4.76	21.16	1
6.85	4.23	-3.47	11.94	1
37.92	48.10	24.99	71.21	1
22.37	9.63	1.71	17.55	0
128.36	120.81	111.17	130.45	1
141.13	80.08	52.48	107.68	0
48.00	44.55	40.47	48.63	1
12.85	-0.37	-28.34	27.61	1
15.62	5.96	-12.64	24.56	1
86.63	121.82	89.02	154.63	0
4.18	6.60	-13.93	27.13	1
18.04	13.76	5.35	22.17	1

Table 4: Marginal post. mode with Gauss kernel (65% validation)

actual	loo_prediction	lower95	upper95	validated
139.51	138.07	134.54	141.60	1
8.09	12.23	8.64	15.82	0
58.93	76.51	59.01	94.02	0
40.77	45.39	40.70	50.08	1
130.14	131.44	128.72	134.15	1
52.37	57.63	49.32	65.94	1
33.72	36.39	25.12	47.66	1
68.25	69.58	63.04	76.12	1
17.53	-9.68	-15.87	-3.50	0
6.85	4.41	-1.76	10.58	1
37.92	65.92	39.47	92.37	0
22.37	8.71	1.32	16.09	0
128.36	124.35	117.21	131.48	1
141.13	101.14	71.38	130.90	0
48.00	45.35	42.00	48.70	1
12.85	-0.23	-25.42	24.95	1
15.62	7.97	-8.08	24.03	1
86.63	120.78	87.49	154.08	0
4.18	8.07	-9.89	26.03	1
18.04	17.07	11.50	22.64	1

Table 5: Marginal post. mode with Matern 5/2 kernel (65% validation)

actual	loo_prediction	lower95	upper95	validated
139.51	138.07	134.54	141.60	1
8.09	12.23	8.64	15.82	0
58.93	76.51	59.01	94.02	0
40.77	45.39	40.70	50.08	1
130.14	131.44	128.72	134.15	1
52.37	57.63	49.32	65.94	1
33.72	36.39	25.12	47.66	1
68.25	69.58	63.04	76.12	1
17.53	-9.68	-15.87	-3.50	0
6.85	4.41	-1.76	10.58	1
37.92	65.92	39.47	92.37	0
22.37	8.71	1.32	16.09	0
128.36	124.35	117.21	131.48	1
141.13	101.14	71.38	130.90	0
48.00	45.35	42.00	48.70	1
12.85	-0.23	-25.42	24.95	1
15.62	7.97	-8.08	24.03	1
86.63	120.78	87.49	154.08	0
4.18	8.07	-9.89	26.03	1
18.04	17.07	11.50	22.64	1

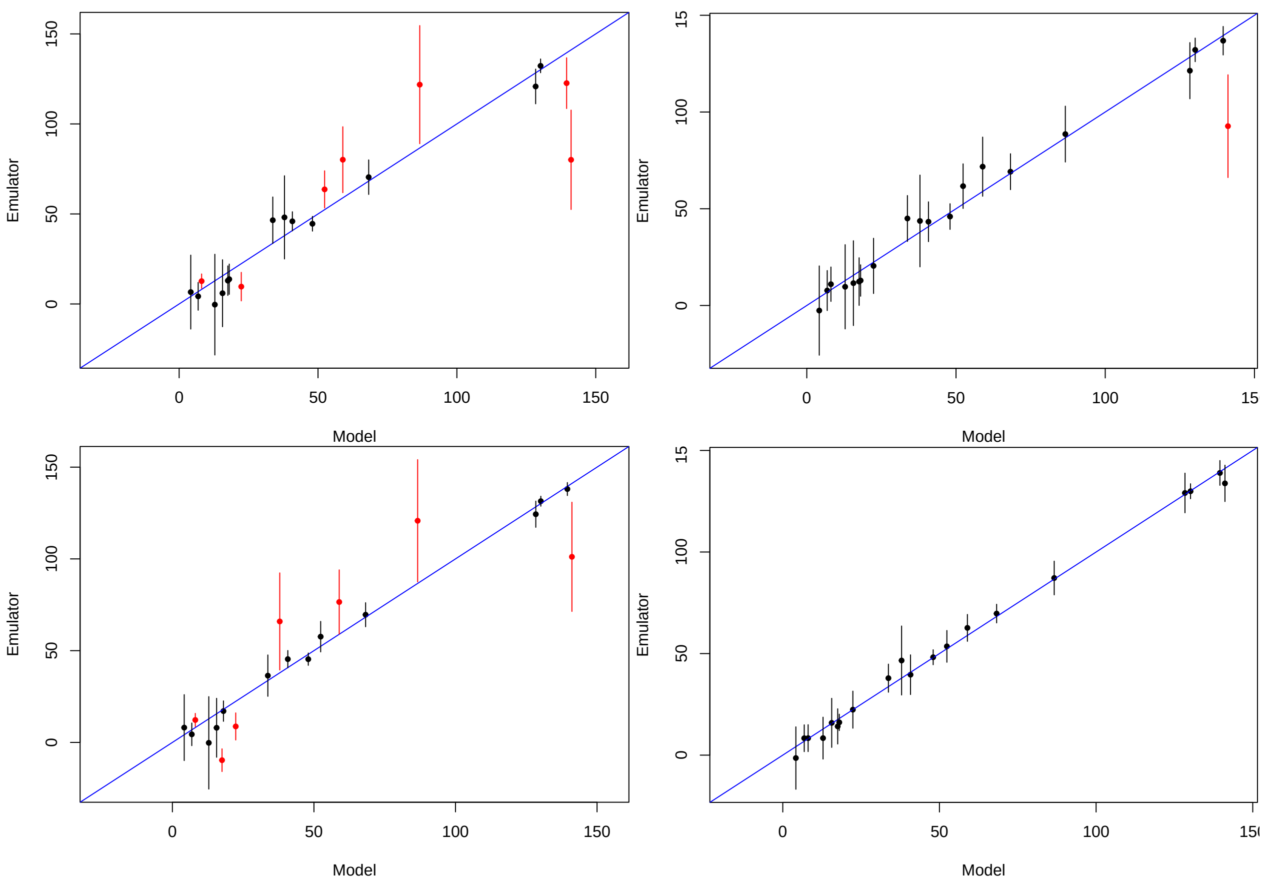


Figure 2: LOO cross validation. Top: Maximum likelihood estimation for the Gauss (left) and Matern 5/2 (right) kernels. Bottom: Marginal posterior mode estimation for the Gauss (left) and Matern 5/2 (right) kernels.

Going forward, for now, only the Gauss kernel is dealt with; Matern 5/2 will be returned to once I've figured out the estimates returned by the two packages and the derivative of the Matern 5/2 kernel. The estimates returned by both estimation methods using the Gauss kernel are shown in Table 6.

Table 6: Estimates table

Estimation	Kernel	Trend 1	Trend 2	Trend 3	Range	Variance
MLE	Gauss	269.52	-128.31	35.43	0.36	21592.3
PM	Gauss	451.46	-23.23	29.93	1.36	119202.48
MLE	Gauss	269.52	-128.31	35.43	0.3	21592.3
PM	Gauss	451.46	-23.23	29.93	1.56	119202.48

Verifying the estimates by-hand for both estimation methods

To calculate the posterior predictions at the 100 testing points by-hand, we require the matrix H containing the inputs for the 20 observations and well as matrix A, its inverse, and $\mathbf{t}(x^*)$ for all 1000 of the new inputs x^*. A and $\mathbf{t}(x^*)$ are built using the estimate of the range parameter returned by each package. By-hand predictions match those returned by the two estimation methods (Figure 4).

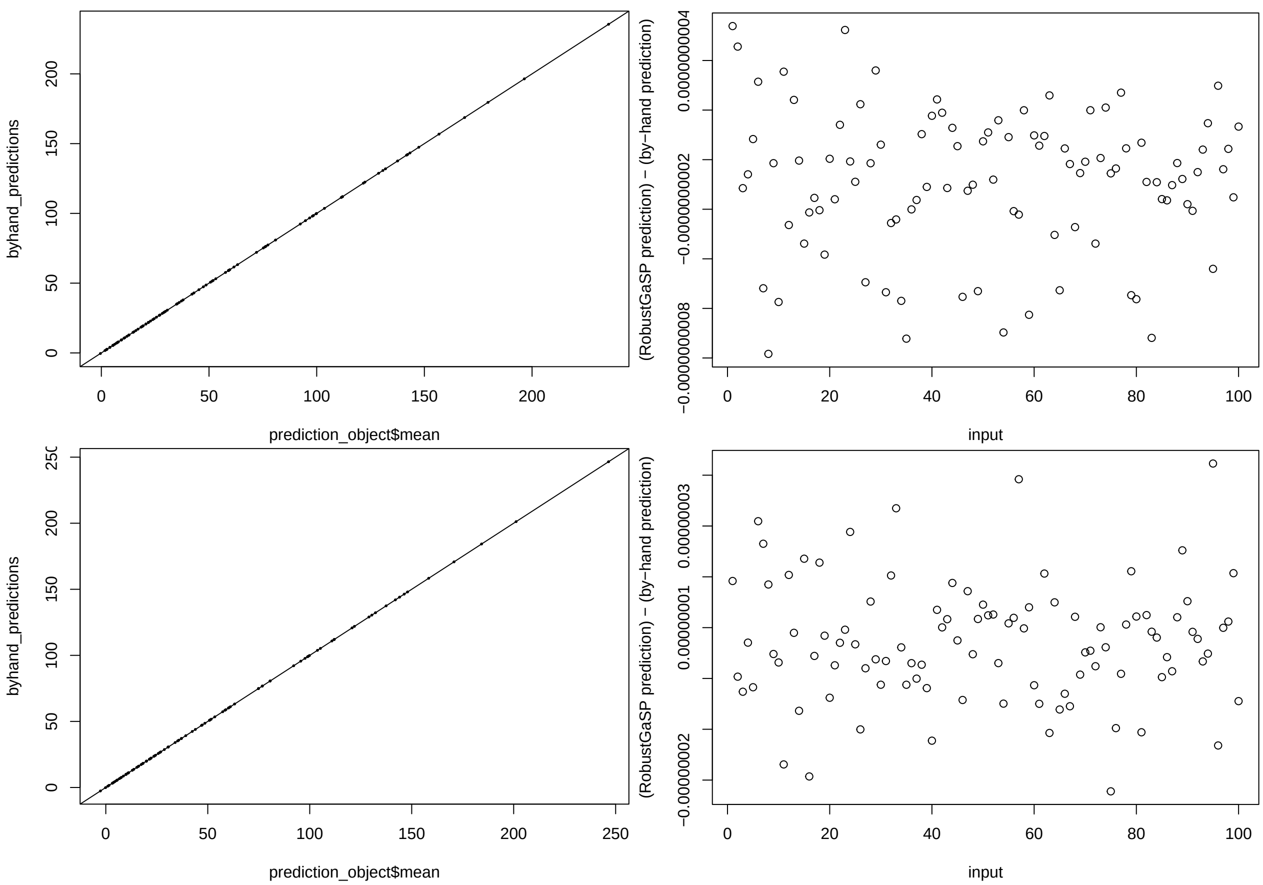


Figure 3: The predictions found by-hand match well with those made by both estimation methods (left), with the latter minus the former found to be essentially zero (right). Top: MLE/ Bottom: RM.

For both we calculate

$$\frac{\sum_{i=1}^{1000} \left| (\text{predictions from package})_i - (\text{predictions by-hand})_i \right|}{1000}$$

then divide this by 10^{-8} (machine precision) to get 0.02 for MLE and 1 for PM (the latter approximately a factor of 47 larger than the former).

Why don't RobustGaSP's estimates change? What is the stochastic element of DiceKriging?

Finally, we create an object containing the normalised partial derivatives for use in the alignment measure.

AM calculation

Finally we calculate the alignment measure of this function with a couple of others. Firstly, this function with itself, for both estimation methods.

```
mean(abs(rowSums(m.mle.gauss.pdsnormed * m.pm.gauss.pdsnormed)))
```

Attempt	AM for ERF and AOD in gb 1	AM for ERF and AOD in gb 2	AM for ERF and AOD in gb 3
1	0.2815566	0.2118547	0.2562991
2	0.2825316	0.2118477	0.2601877
3	0.2652177	0.2117447	0.2601681
4	0.2647665	0.211873	0.2197906
5	0.2653364	0.2118245	0.2834103
6	0.2821098	0.1850545	0.2906794
7	0.2653951	0.2040215	0.2541661
8	0.2648363	0.2118733	0.2600939
9	0.2653367	0.1116973	0.2562923
10	0.2701372	0.2118717	0.2599827
Robust	0.01612035	0.2331373	0.01426525